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Iris-3B: Going Beyond the Latent with Pixel-Space Diffusion Training, Conversion and Fine-Tuning

A 3B-Parameter VAE-Free Diffusion Model Matches Latent Baselines on Text-to-Image Quality but Finds No Advantage on Fine-Grained Pixel Tasks

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Iris-3B is a 3-billion-parameter text-to-image diffusion transformer trained entirely in pixel space—no VAE encoder, no latent compression—via a 256→512→1024 resolution curriculum, achieving GenEval 0.798 and DPG-Bench 86.52, matching latent FLUX.2 Klein. Fine-tuning on depth estimation and super-resolution finds no improvement over the latent baseline, suggesting the VAE bottleneck is less limiting than assumed.

AT A GLANCE

Problem: VAE bottleneck in latent diffusion may limit pixel-level task quality; direct pixel-space diffusion is a natural alternative.

Why it matters: If pixel-space diffusion is competitive, the field can revisit the latent paradigm assumption.

Method: PiT pixel-space diffusion transformer, 3B parameters, trained with a 256→512→1024 resolution curriculum.

Result: GenEval 0.798, DPG-Bench 86.52; no gain on depth estimation or 4× super-resolution over latent baselines.

Evidence: Compared against latent FLUX.2 Klein fine-tunes on downstream tasks; converted pixel model underperforms both.

The Big Picture

The dominant paradigm in text-to-image generation encodes images into a latent space with a VAE before denoising with a diffusion model. This reduces compute: a 1024×1024 image is processed as a ∼128×128 latent, cutting memory and FLOPs by up to 64×. But the VAE reconstruction introduces artefacts—blurring, ringing, and colour shift—that may degrade performance on pixel-level tasks such as depth estimation and super-resolution.

Iris-3B asks: does removing the VAE bottleneck and training directly in pixel space yield measurable gains on these tasks? The answer, after a thorough empirical investigation, is: not yet.

Why Existing Approaches Fall Short

Latent diffusion models (LDMs) incur VAE reconstruction error at inference time. For generation quality benchmarks, this error is tolerated; for fine-grained pixel tasks, it may be a genuine bottleneck.

Pixel-space diffusion at high resolution is expensive: the denoising network must process full-resolution feature maps, scaling quadratically in spatial dimensions.

Cascaded super-resolution (used in early pixel-space models like Imagen) adds architectural complexity and requires training separate upsampling stages.

Latent-to-pixel conversion (fine-tuning a pre-trained latent model in pixel space) is unexplored as an alternative to training from scratch.

Core Mechanism

Iris-3B (from scratch): A PiT (Pixel-space Image Transformer) diffusion model with 3B parameters. Training uses a progressive resolution curriculum: stage 1 at 256^2, stage 2 at 512^2, stage 3 at 1024^2. Text conditioning uses Qwen3-VL-4B-Instruct as the text encoder.

Converted pixel model: FLUX.2 Klein 4B is adapted to pixel space by replacing the VAE encoder/decoder with an identity mapping and adjusting channel dimensions, then fine-tuned on the pixel-space denoising objective.

Both are then fine-tuned on downstream tasks (monocular depth, 4× super-resolution).

Technical Formulation

The pixel-space denoising objective is standard denoising score matching:

$$\mathcal{L}_{\text{DSM}} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\Big[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})\|_2^2\Big]$$

where $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}$ is the noisy image at timestep t and $\mathbf{c}$ is the text embedding.

The progressive training schedule uses a staged sequence $(256^2, T_1) \to (512^2, T_2) \to (1024^2, T_3)$ with decreasing step counts at higher resolutions to avoid excessive compute at the final stage.

For task-specific fine-tuning (e.g., depth estimation with ground-truth $\mathbf{d}$):

$$\mathcal{L}_{\text{task}} = \mathcal{L}_{\text{DSM}}(\mathbf{d}) + \lambda\|\hat{\mathbf{d}} - \mathbf{d}\|_1$$

Learning or Inference Procedure

Pre-training:

1. Stage 1: train at 256^2 on image–caption pairs for T_1 steps.
2. Stage 2: upsample, continue at 512^2 for T_2 steps.
3. Stage 3: continue at 1024^2 for T_3 steps.

Fine-tuning: Replace conditioning input with the task signal; train with the task-specific loss for a small number of steps.

Inference: Standard DDPM/DDIM sampling; no VAE decoding.

What the Guarantee Says

No theoretical guarantees are claimed. The paper makes empirical claims on GenEval and DPG-Bench (generation quality) and on monocular depth and DIV2K super-resolution (fine-grained tasks).

Experimental Findings

- **Generation quality:** Iris-3B achieves **GenEval 0.798** and **DPG-Bench 86.52** at 1024^2, matching Qwen-Image and competitive with latent FLUX.2 Klein.
- **Monocular depth:** Iris-3B performs at parity with a latent FLUX.2 Klein fine-tune; the converted pixel model performs worse.
- **4× super-resolution (DIV2K):** Neither Iris-3B nor the converted model outperforms the latent FLUX.2 Klein fine-tune.
- **Inference cost:** Pixel-space at 1024^2 is substantially more expensive per step than latent inference.

Ablations and Interpretation

Curriculum stages: Skipping the 256^2 stage degrades final generation quality; low-frequency pre-training is important for convergence.

Conversion quality: The converted pixel model underperforms both the from-scratch model and the original latent model, suggesting that the latent representation carries useful inductive bias.

Text encoder: Replacing Qwen3-VL-4B with a smaller encoder degrades text alignment (DPG) more than object-count metrics (GenEval).

Interpretation: The null result suggests that either the VAE bottleneck is less harmful than assumed, that the latent representation is beneficial beyond compression, or that pixel-space models need more task-specific data to close the gap.

Reference

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